Executive Summary – KKBox Churn Prediction

Project Overview

This project predicts user churn for KKBox, a music streaming service, using the public WSDM KKBox Music Recommendation Challenge dataset. The dataset spans January 2015 to February 2017 and includes transaction history, user demographics, and listening logs. We developed an end-to-end pipeline from raw data to a deployable model that identifies users at risk of churning. Users are labeled as churned if they fail to renew their subscription within 30 days of expiration.

Data Pipeline

We generated 25 monthly cohorts using custom transaction sorting logic (Notebook 1) and extracted listening "velocity" features (Notebook 2). Exploratory analysis (Notebook 3) revealed that auto-renewal status, recency of listening, and cancellation history are the strongest predictors of churn. In Notebook 4, we merged all sources, recreated ratio_auto_renew from Notebook 3, imputed missing values using train-only statistics, and split the data into:

- Train: Feb 2015 – Sep 2016 (20 cohorts, ~12.6M rows)
- Validation: Oct – Nov 2016 (2 cohorts, ~1.7M rows)
- Test: Dec 2016 – Feb 2017 (3 cohorts, ~2.6M rows)

Modeling and Results

We first trained five models (Logistic Regression, Random Forest, XGBoost, LightGBM, and CatBoost) on the 20 training cohorts and used expanding-window time-series cross-validation (Notebook 5) to assess performance stability over time. We then evaluated all five models on the two validation cohorts. The final metric we decided to use for model selection is PR_AUC since we have an imbalanced dataset, with much more non-churn than churn. XGBoost achieved the best validation PR-AUC of 0.759. Hyperparameters were tuned using Bayesian optimization. Then, training XGBoost with the tuned parameters on the train and validation cohorts and testing on the three unseen months in the test set resulted in an ROC-AUC: 0.83, log loss: 0.148, and PR-AUC: 0.542.

We also tried an ensemble model (Notebook 6) weighting the five models based on their performance on the time-series cross-validation PR-AUC. After validating the ensemble model on the two validation cohorts, we only saw a modest increase to 0.761, not justifying the added complexity. XGBoost offers comparable performance with better interpretability, making it the preferred choice for business stakeholders. So, we selected XGBoost as our final model.

Beyond prediction, we built an LLM-powered explanation layer that translates the model's per-user SHAP attributions into plain-language churn reports for retention teams (Notebook 7). We focus on describing the features that led to the label and the direction in which they influenced it. The pipeline works as follows:

1. SHAP computes per-feature contributions for each user.
2. A structured extractor ranks top drivers (e.g., "last transaction was a cancellation," "recency of use").
3. A small, free open-source LLM (Qwen2.5-1.5B) generates a business-readable report summarizing the SHAP analysis and suggesting retention actions.

Crucially, the LLM never invents reasons; every factual claim comes from SHAP, making the report auditable and reliable. This layer bridges the gap between technical model outputs and business decision-making, enabling retention teams to understand not just who is at risk, but what is driving it.

Implications and Business Value

The model provides actionable insights for customer retention:

1. **Top Predictive Features** (from XGBoost):
- *last_is_auto_renew*: Users whose last transaction was an auto-renewal are significantly less likely to churn. Promoting auto-renewal sign-up is a high-impact retention lever.
- *last_is_cancel*: A cancellation in the last transaction is the strongest churn signal. These users should be flagged immediately for targeted win-back campaigns.
- *days_since_last_use*: Recency of listening is a leading indicator of engagement. Users inactive for 7+ days should receive re-engagement notifications.

2. **Operational Recommendations**:
- *Auto-Renewal Campaigns*: Target users who have never enabled auto-renewal with incentives (e.g., a free month) to possibly reduce churn.
- *Re-Engagement Triggers*: Automatically send personalized content recommendations to inactive users.
- *Cancellation Alerts*: Build a real-time flag for users who cancel their subscription, triggering a retention offer (e.g., discount, exclusive content).

3. **Model Deployment**:
The model is lightweight enough for batch scoring (e.g., nightly jobs) and can be integrated into a customer relationship management system to generate a daily "at-risk" user list for retention teams.

4. **LLM Reports for Retention Teams**:
The SHAP-to-LLM pipeline can be run on high-risk users to produce actionable, human-readable summaries, reducing the time retention specialists spend interpreting model outputs.

5. **Future Improvements**:
- *Real-Time Scoring*: Extend the pipeline to score users daily based on recent activity.
- *Feature Expansion*: Incorporate additional data sources (e.g., playlist interactions, social features) to improve PR-AUC.
- *Transaction Collection*: Implement a better transaction collection method for the team so that we would not have to drop the customers with more than 2 plan transactions in a day (a small number of customers) to properly sort transactions for the churn labeling process.
- *Member Demographics*: Incorporate member demographic data. The member demographics provided had high missingness, and the age lacked enough specificity.
- *Usage Velocities*: Explore different usage velocity windows. The current usage velocities are calculated by dividing 14-day usage by 30-day usage relative to the cohort cutoff date. Using nonoverlapping windows could provide more powerful usage features.
- *Muddy Features*: Separate the strong signals better using refined features.
    - *avg_plan_price*: receives mixed price signals from users on expensive plans and those on longer contracts.
    - *avg_payment_per_day*: the calculation method causes a low value for users who are returned past churners. Separating the signal coming from past churn and cost might produce a better-performing model.

Conclusion

This project delivers a robust, interpretable churn-prediction model that identifies at-risk users and explains why they are at risk. XGBoost was selected as the final model due to its strong performance and interpretability. The LLM explanation layer further enhances business value by translating technical attributions into clear, actionable reports. The top drivers: auto-renewal status, recency of use, and cancellation history provide a clear playbook for retention teams. By acting on these signals, KKBox can proactively reduce churn and improve customer lifetime value.